

Shriya Patel

Software Engineer

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SUMMARY:

Software Engineer with 4+ years of experience building **scalable backend systems, microservices**, and **cloud-native applications** using **Python, Java, FastAPI, Spring Boot**, and **Node.js**. Strong expertise in **distributed systems, REST/gRPC APIs, event-driven architectures**, and enterprise applications deployed on **AWS** and **Kubernetes**. Experienced in designing high-performance systems with **Docker, Kafka, Redis, CI/CD pipelines**, and database optimization. Proven ability to improve application scalability, reliability, and performance through modern engineering practices. Passionate about **backend engineering, cloud computing**, and building resilient distributed platforms for enterprise-scale applications.

TECHNICAL SKILLS:

Programming Languages: Python, Java, C++, C#, SQL, JavaScript, TypeScript

Backend Development & Frameworks: FastAPI, Spring Boot, Django, Node.js, Express.js, .NET Core, REST APIs, gRPC, Microservices Architecture, System Design

Frontend Technologies: React.js, Next.js, HTML5, CSS3

Cloud & Distributed Systems: AWS (Lambda, EC2, S3, SNS, SES, EKS), Azure (Functions, AKS, Blob Storage, Service Bus), Distributed Systems, Event-Driven Architecture

DevOps & CI/CD: Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps, CI/CD Pipelines, Linux, Bash

Databases & Caching: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis

Data Streaming & Big Data: Apache Kafka, Apache Spark, Hadoop

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, NLP, Computer Vision, MLflow

Security & Authentication: OAuth2, JWT, RBAC, TLS, AES Encryption

Monitoring & Observability: Prometheus, Grafana, ELK Stack, CloudWatch, Logging & Metrics

Tools & Version Control: Git, GitHub, Maven, Gradle, Postman, VS Code, IntelliJ IDEA

PROFESSIONAL EXPERIENCE:

ServiceNow, USA | Software Engineer | Nov 2024 - Present

- Architected and deployed **scalable microservices** using **Python (FastAPI)** and **Node.js** on **AWS Lambda** and **Kubernetes**, improving system scalability and supporting high-volume enterprise workflows.
- Designed high-performance **REST** and **gRPC APIs** integrated with **PostgreSQL** and **Redis**, reducing API response latency by **32%** through caching and query optimization.
- Implemented **event-driven communication** using **Apache Kafka** and **AWS SNS**, reducing inter-service bottlenecks by **28%** and improving real-time processing reliability.
- Automated containerized deployments using **Docker**, **Amazon EKS**, and **CI/CD pipelines**, decreasing release failures by **25%** and accelerating deployment cycles.
- Strengthened application security by implementing **OAuth2, JWT**, and **RBAC authentication** mechanisms across distributed systems.
- Built centralized **observability** and monitoring pipelines with **Kubernetes logging**, metrics, and troubleshooting dashboards, improving incident detection efficiency.

Disha Enterprise, India | Software Engineer | Jan 2021 - Jul 2023

- Developed a cloud-based **log analytics platform** using **C#/.NET Core** and **C++**, processing over **1M+ log events daily** with optimized backend performance.
- Engineered scalable backend services using **Spring Boot** and **.NET Core** integrated with **MySQL** and **MongoDB** for high-volume enterprise applications.
- Built real-time streaming pipelines using **Apache Kafka**, increasing system throughput by **30%** and improving data processing efficiency.
- Developed responsive frontend components using **React** and **TypeScript**, improving application usability and user engagement by **20%**.
- Optimized **PostgreSQL** and **DynamoDB** database queries, reducing execution time by **35%** and improving application responsiveness.
- Designed **multithreaded C++ modules** for high-speed log parsing and caching, enhancing processing efficiency for large-scale datasets.

PROJECTS:

Air Pollution Monitoring System

- Built a fault-tolerant **IoT data pipeline** processing **2M+ MQTT messages daily** using **Azure IoT Hub** and **AWS cloud services**.
- Developed scalable **microservices** for real-time environmental monitoring and anomaly detection with high system reliability.
- Improved platform **observability** and maintained **99% uptime** through monitoring, logging, and resilient cloud architecture.

Real-Time Chat Application

- Developed a scalable **real-time chat application** using **React, Node.js**, and **Socket.io** supporting **100+ concurrent users**.
- Implemented secure user authentication and session management using **JWT** and **bcrypt encryption**.
- Improved application scalability and message delivery performance using **Redis caching** and **Azure Service Bus** integration.

3D Object Detection System

- Built high-accuracy **object detection models** using **YOLO, TensorFlow**, and **PyTorch**, achieving up to **95–100% prediction accuracy**.
- Developed scalable **machine learning pipelines** for model training, optimization, and deployment in production environments.

EDUCATION:

Master of Science in Information Technology | Arizona State University, USA • Graduated May 2025 | **GPA: 4.0/4.0**

Bachelor of Engineering in Computer Science | Indus University, India • Graduated May 2023 | **GPA: 3.96/4.0**

CERTIFICATE:

[Microsoft Azure AI Essentials Professional](#)

[Level up LLM applications development with LangChain and OpenAI](#)

PUBLICATIONS:

- Solar Radiation and Solar Energy Estimation using ANN and Fuzzy Logic Concept (2022)**
DOI: <https://doi.org/10.1007/s11356-022-19185-z>.
- Artificial Intelligence and Cyber Security in Cryptocurrency and Banking Systems (2022)**
DOI: <https://link.springer.com/article/10.1007/s40745-022-00433-5>.